TEACH YOURSELF INTERMEDIATE XI UNIT 1

Computer Systems

Digital logic, SDLC, and networking foundations

What you will master

- Differentiate analog vs digital representation and signals
- Use Boolean algebra, truth tables, and logic gates (including Logisim)
- Compare Waterfall and Agile SDLC with case studies
- Explain OSI and TCP/IP models and layer protocols

How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

1.1 Discrete vs continuous & digital systems

Discrete quantities are countable (students in a class, books on a shelf). **Continuous** quantities vary smoothly (temperature, height, time).

Stairs = discrete

Only fixed step heights — like binary 0/1 levels.

Ramp = continuous

Any height in between — like analog voltage.

A **digital system** uses bits: (OFF/LOW/FALSE) and (ON/HIGH/TRUE). Digital systems resist noise, copy accurately, and integrate with software.

Remember
Modern phones, calculators, and traffic lights are digital at their core.

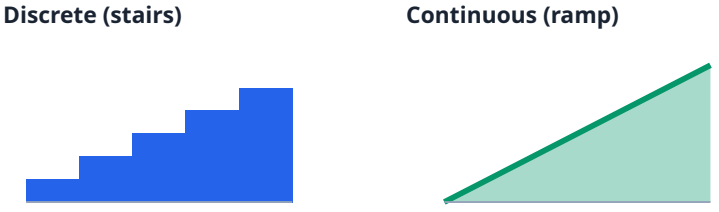


Figure 1.1 — Discrete steps vs continuous slope (analog vs digital intuition).

1.2 Analog vs digital signals

A **signal** carries information. **Analog** signals vary continuously (voice, old radio). **Digital** signals switch between HIGH and LOW (computer data, USB, internet packets).

Feature	Analog	Digital
Values	Infinite in range	Two levels (0, 1)
Wave shape	Smooth (sine-like)	Square steps
Noise	Easily distorted	Easy to regenerate
Examples	Vinyl, mercury thermometer	PC, smartphone, digital watch

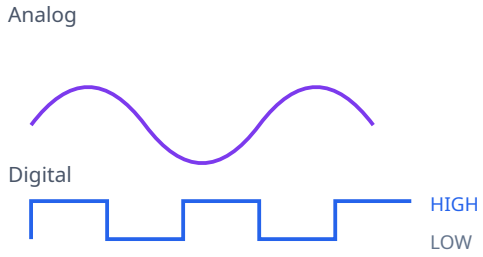


Figure 1.2 — Smooth analog wave vs two-level digital wave.

1.3 Boolean algebra & operations

Boolean algebra (George Boole) uses only and . Variables hold binary values; expressions combine them with AND, OR, NOT.

AND ( ): output only if *all* inputs are . Truth table rows for inputs: .

A	B	
0	0	0
0	1	0
1	0	0
1	1	1

OR ( ): if *any* input is . **NOT** ( ): flips the bit.

XOR ( ): when inputs differ: .

XNOR: when inputs are equal. **NAND** and **NOR** are *universal gates* — any circuit can be built from them alone.

> **Exam tip**
> For 3 inputs, list binary combinations from 000 to 111 in order when building truth tables.



Figure 1.3 — Standard logic gate symbols (exam drawing practice).

1.4 Logisim & building circuits

Use **Logisim Evolution** to place gates, input pins (square), output pins (circle), and orthogonal wires. Simulate all input combinations.

Example: 2-input AND — only when and . Wire colors: bright green = 1, dark green = 0.

> **Try it**
> Build XOR from AND, OR, and NOT. Verify with four test cases through .

1.5 Software Development Life Cycle (SDLC)

SDLC is the structured path from idea to working software: requirements → design → coding → testing → deployment → maintenance.

Waterfall — phases run sequentially; best when requirements are stable (e.g. school library system with fixed rules).

Agile (Scrum) — short *sprints* (often 2 weeks), evolving product backlog, frequent demos. Fits mobile apps with changing stakeholder needs (timetable, attendance, notifications).

Agile artifacts: product backlog, user stories, sprint reviews, retrospectives.

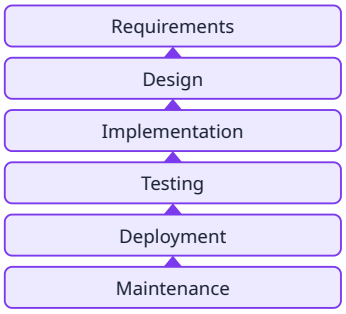


Figure 1.4 — Waterfall: each phase completes before the next begins.

1.6 OSI & TCP/IP

The **OSI model** has 7 layers — a teaching map for network functions. Memorize top-to-bottom:

1. Physical — bits on wire/fiber
2. Data Link — frames, MAC addresses
3. Network — routing, IP addresses
4. Transport — TCP/UDP, ports, reliability
5. Session — dialog control
6. Presentation — encoding, encryption
7. Application — HTTP, email, DNS to users

TCP/IP (4 layers) is what the real internet uses: Network Access, Internet, Transport, Application. TCP gives reliable delivery; UDP is faster but unreliable.

> **Remember**
> "All People Seem To Need Data Processing" — Application (7) down to Physical (1).

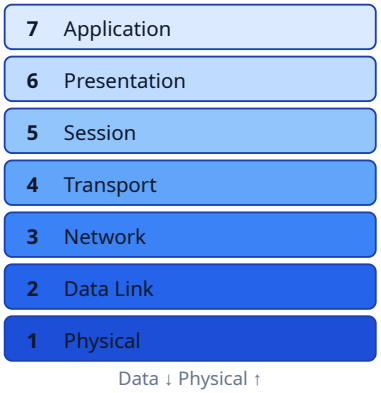


Figure 1.5 — OSI stack: each layer serves the layer above.

Teach Yourself — Practice & Check

1. Draw truth tables for NAND, NOR, and 3-input AND.
2. Simplify using Boolean laws and build in Logisim.
3. List two reasons a school might choose Agile over Waterfall for a student app.
4. Name one protocol or device at OSI layers 2, 3, and 7.